

Yijun Liang

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Ph.D. Student, Department of Computer Science, University of Maryland, College Park

Research Interests: Computer Vision, Visual Perception & Reasoning, Data-Centric AI (Data Synthesis, Benchmark Design)

EDUCATION

- **University of Maryland, College Park** Aug 2023 – Present
Ph.D. in Computer Science. Maryland, U.S.
- **University of California, Berkeley** Aug 2019 – Dec 2020
M.Eng. in Industrial Engineering & Operations Research Berkeley, U.S.
- **Tsinghua University** Aug 2015 – Jul 2019
B.E. in Industrial Engineering (Operations Research) Beijing, China

PUBLICATIONS

- Visual Contrastive Self-Distillation** [\[arXiv\]](#) arXiv 2026
Yijun Liang, Yunjie Tian, Yijiang Li, Yuqi Jia, Furong Huang, Tianyi Zhou, Di Fu
- Self-Evolving Visual Questioner** [\[arXiv\]](#) arXiv 2026
Yijun Liang, Hengguang Zhou, Ming Li, Lichen Li, Cho-Jui Hsieh, Tianyi Zhou
- LLMs Struggle to Measure What Distinguishes Students of Different Proficiency Levels: A Study of Item Discrimination in Reading Comprehension Assessment** [\[arXiv\]](#) arXiv 2026
Han Chen, Ming Li, Chenguang Wang, Yijun Liang, Dawei Zhou, Hong Jiao, Tianyi Zhou
- Do Prompt-Elicited Trajectories Reflect Training-Time Reward Hacking? A Systematic Study on Monitoring Training-Time Reward Hacking in Code Generations** [\[arXiv\]](#) arXiv 2026
Lichen Li, Hengguang Zhou, Yijun Liang, Tianyi Zhou, Cho-Jui Hsieh
- V-REX: Benchmarking Exploratory Visual Reasoning via Chain-of-Questions** [\[arXiv\]](#) ECCV 2026
Chenrui Fan*, Yijun Liang*, Shweta Bhardwaj*, Kwesi Cobbina, Ming Li, Tianyi Zhou
- History-Conditioned Spatio-Temporal Visual Token Pruning for Efficient Vision-Language Navigation** [\[arXiv\]](#) IROS 2026
Qitong Wang, Yijun Liang, Ming Li, Tianyi Zhou, Christopher Rasmussen
- ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness** [\[arXiv\]](#) NeurIPS 2025
Yijun Liang*, Ming Li*, Chenrui Fan, Ziyue Li, Dang Nguyen, Kwesi Cobbina, Shweta Bhardwaj, Jiuhai Chen, Fuxiao Liu, Tianyi Zhou
- Diffusion Curriculum: Synthetic-to-Real Generative Curriculum Learning via Image-Guided Diffusion** [\[arXiv\]](#) ICCV 2025
Yijun Liang*, Shweta Bhardwaj*, Tianyi Zhou
- Mosaic-IT: Free Compositional Data Augmentation Improves Instruction Tuning** [\[pdf\]](#) ACL 2025
Ming Li, Pei Chen, Chenguang Wang, Hongyu Zhao, Yijun Liang, Yupeng Hou, Fuxiao Liu, Tianyi Zhou
- CaughtCheating: Is Your MLLM a Good Cheating Detective? Exploring the Boundary of Visual Perception and Reasoning** [\[arXiv\]](#) arXiv 2025
Ming Li, Chenguang Wang, Yijun Liang, Xiyao Wang, Yuhang Zhou, Xiyang Wu, Yuqing Zhang, Ruiyi Zhang, Tianyi Zhou
- PEDANTS: Cheap but Effective and Interpretable Answer Equivalence** [\[arXiv\]](#) EMNLP 2024
Zongxia Li, Ishani Mondal, Yijun Liang, Huy Nghiem, Jordan Lee Boyd-Graber
- GeoDRL: A Self-Learning Framework for Geometry Problem Solving Using Reinforcement Learning in Deductive Reasoning** [\[pdf\]](#) ACL 2023
Shuai Peng, Di Fu, Yijun Liang, Liangcai Gao, Zhi Tang
- Compute Like Humans: Interpretable Step-by-Step Symbolic Computation with Deep Neural Network** [\[pdf\]](#) KDD 2022
Shuai Peng, Di Fu, Yong Cao, Yijun Liang, Gu Xu, Liangcai Gao, Zhi Tang

WORK EXPERIENCE

- **Research Scientist Intern**
TikTok, Seattle, WA
 - **Image & Video VLM post-training:** Developed on-policy distillation (OPD) and on-policy self-distillation (OPSD) recipes for image and video vision–language models.
 - **Privileged-information distillation:** Studied training-time privileged views available only to the teacher, replacing uniform KL with per-token weighting so the student is supervised on signal it can ground.
 - **Research Scientist Intern**
TikTok, San Jose, CA
 - **Data synthesis:** Generated synthetic videos to improve temporal reasoning and motion understanding in VideoLLMs.
 - **Video Encoder Optimization:** Refined the masking module to eliminate redundancy and reduce token usage for efficient motion-aware understanding.
 - **Machine Learning Engineer**
TikTok, Beijing, China
 - **Video Violation Detection with VLMs:** Developed a vision–language moderation framework to identify infrequent or long-tail policy violations in video data.
 - **Video–Text Classification & Generation:** Developed DeBERTa-based classifiers and generators for multi-video Q&A, scaling content generation to over 100M high-quality pairs.
 - **Symbolic Computation Models:** Designed transformer architectures with novel Huffman embeddings and sub-tree pretraining to improve mathematical problem-solving.

May 2026 – Aug 2026

May 2024 – Aug 2024

Mar 2021 – Jul 2023

HONORS & AWARDS

- **Silver Reviewer Award, ICML**
 - **Student Conference Travel Award, ICCV**
 - **Dean’s Fellowship and Chair’s Fellowship Award, University of Maryland, College Park**
- 2026

2025

2023

ACADEMIC SERVICE

- **Program Committee, AAAI**
 - **Reviewer, ICML**
- 2026

2026